

# MUHAMMAD MUSTAFA

Software Engineer | AI/ML Engineer | Web Developer

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## PROFESSIONAL SUMMARY

Computer Science student with a strong foundation in Software Engineering, Artificial Intelligence, Machine Learning, and Full-Stack Web Development. Experienced in building scalable applications, designing REST APIs, applying data structures and algorithms, and deploying intelligent systems. Proven leadership through ACM and Procom roles. Passionate about developing efficient, real-world software solutions across AI/ML, backend, and frontend domains.

## TECHNICAL SKILLS

**Programming Languages:** C, C++, Python, JavaScript, TypeScript, HTML, SQL

**Frontend Development:** React.js, Tailwind CSS, Responsive Design, Vite

**Backend Development:** Node.js, Express.js, REST APIs

**Databases:** PostgreSQL, MongoDB, Oracle SQL, Supabase

**AI / ML:** Machine Learning, A\* Search, CSP, Dijkstra, NetworkX, Scikit-learn

**Tools & Practices:** Git, GitHub, OOP, Design Patterns, Agile, Streamlit

**Languages:** English (Fluent), Urdu (Native)

## EDUCATION

### Bachelor of Science in Computer Science

FAST National University of Computer and Emerging Sciences (FAST-NUCES), Karachi | Aug 2023 – Jun 2027

CGPA: 3.05 / 4.0

## WORK EXPERIENCE

### Teaching Assistant (TA) — FAST-NUCES, Karachi

Aug 2025 – May 2026

- Supported 100+ students across Linear Algebra, Calculus, and Database Systems courses.
- Provided targeted guidance on debugging, algorithm analysis, and structured problem-solving strategies.
- Evaluated assignments and lab work, delivering actionable feedback to improve student outcomes.

## TECHNICAL PROJECTS

### Algo Rank — Competitive Programming Platform

Technologies: React.js, Node.js, Express.js, PostgreSQL, REST APIs

- Architected a real-time coding competition platform with dynamic leaderboards and live scoring.
- Implemented secure JWT-based authentication, problem submission engine, and concurrent session management.
- Delivered a full-stack solution supporting 200+ competition participants with scalable backend services.

### Karachi Market Intelligent Path Finder

Technologies: Python, Streamlit, A\* Search, UCS Algorithm, API Integration

- Developed an AI-based navigation system leveraging A\* and Uniform Cost Search on real geographical data.
- Optimized pathfinding algorithms achieving 40% improvement in computation speed over baseline approaches.
- Integrated live map APIs and deployed via Streamlit for an interactive user experience.

## Graph-Based Shortest Path Analyzer

*Technologies: Python, Machine Learning, Dijkstra's Algorithm, NetworkX, Scikit-learn*

- Implemented optimized shortest-path computation using Dijkstra's algorithm on weighted graph networks.
- Trained a supervised ML model achieving 85% accuracy in travel time prediction using feature-engineered datasets.

## Enterprise Bank Management System

*Technologies: C++, OOP, Data Structures, Design Patterns*

- Built a full-featured banking system supporting account management, fund transfers, and loan processing.
- Applied OOP principles and design patterns (Factory, Singleton) for a modular, maintainable codebase.

## Air Ticket Reservation System

*Technologies: React.js, Node.js, Express.js, Oracle SQL*

- Developed a flight booking platform with real-time seat availability, search filters, and payment integration.
- Designed normalized Oracle SQL schema and REST APIs supporting concurrent booking transactions.

## LEADERSHIP & ACTIVITIES

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### Co-Head – Extended Procom (Media Production) | FAST-NUCES | 2025–2026

- Led a 10+ member team in multimedia content creation for Procom, Pakistan's largest computing event.
- Supervised end-to-end design, animation, and video production for a national-level competition.

### Deputy Organizer – AI Competitions (Vibe Coding) | Procom | 2026

- Organized AI-focused coding competitions attracting 200+ participants from universities nationwide.
- Managed event logistics, problem design, judging criteria, and full-cycle coordination.

### Coordinator – ACM Animation Team | ACM FAST Chapter | 2024–2025

- Produced and edited 15+ promotional videos for university events, workshops, and campaigns.
- Delivered high-quality multimedia projects consistently under tight deadlines.

## ACHIEVEMENTS & CERTIFICATIONS

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- Star Performer Award — ACM FAST Chapter (Technical Contribution & Leadership)
- Shields Award — Procom FAST (Event Management & Team Coordination)
- Star Performer Award — Procom FAST (Leadership & Technical Excellence)
- Multiple Competition Certificates — Procom, DevDay, Coders Cup (2024–2026)